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Lecture 10A

Before-Tutorial-Session

Tut08

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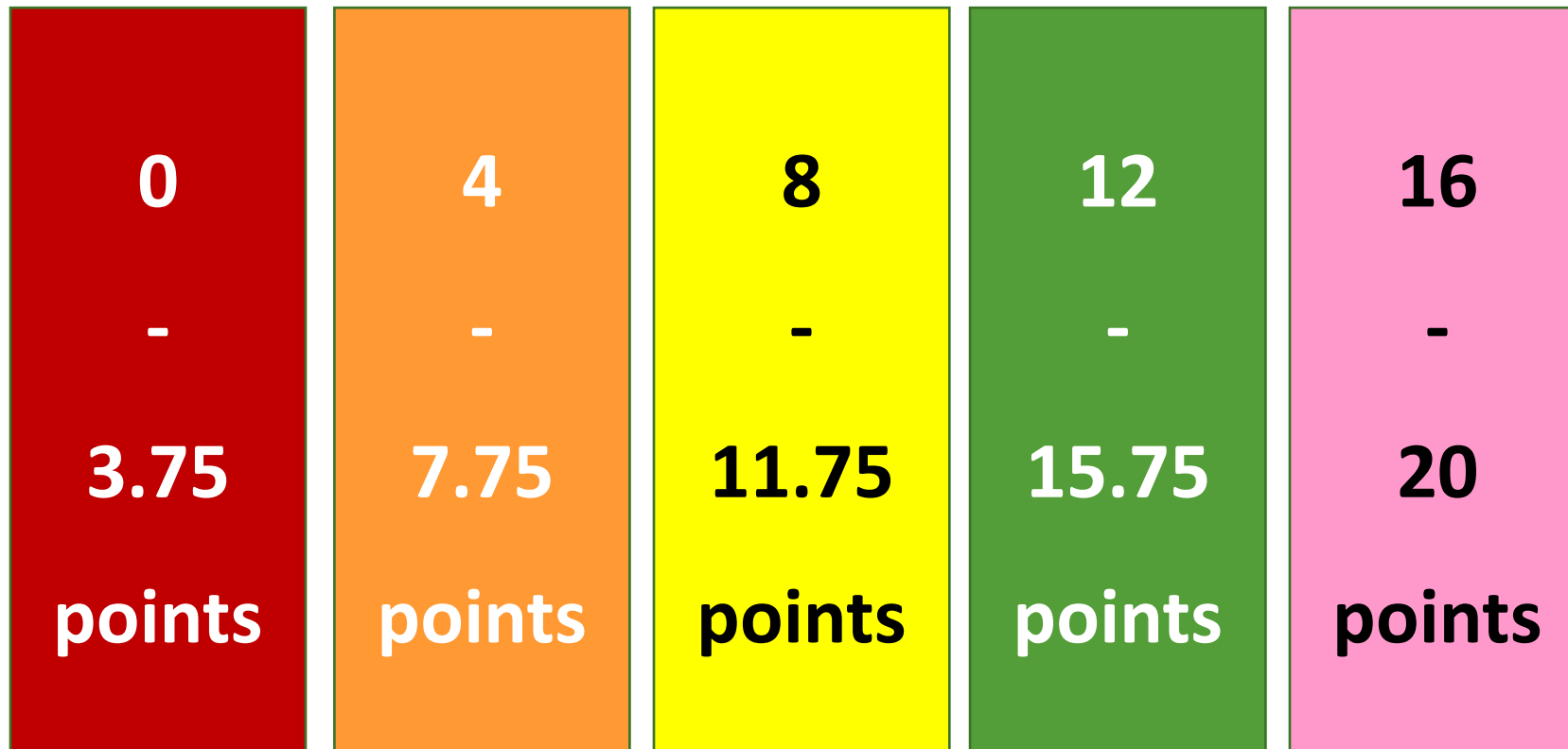
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The Wooclap scores are available on NTULearn.

Which box do you belong to?

To obtain one point, you need to answer TRUTHFULLY.



Recall:

Diagonalization Theorem.

Let A be a real $n \times n$ -matrix. Then TFAE.

(I) We have that A is diagonalizable.

(III) We have that

$\alpha(\lambda_i) \leftarrow$ alg. multiplicity
 $\dim(E_{\lambda_i}) \leftarrow$ geom. multiplicity
 $\uparrow$
eigenspace

$$\alpha(\lambda_i) = \dim(E_{\lambda_i}) \text{ for each eigenvalue } \lambda_i \quad \text{and} \quad \dim(E_{\lambda_1}) + \dim(E_{\lambda_2}) + \dots + \dim(E_{\lambda_s}) = n$$

invertible
diagonal

$$\hat{A} = \hat{P} \hat{D} \hat{P}^{-1}$$

Recall:

Proposition (geometric is at most algebraic).

Let A be a real $n \times n$ -matrix and let λ_i be an eigenvalue of A .

Then

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$$1 \leq \dim(E_{\lambda_i}) \leq \alpha(\lambda_i)$$

$\downarrow$ geometric $\downarrow$ algebraic

Let's do these questions today. Tut08, Q4+Q8

Question 4. For each of the following real matrices, find its real eigenvalues. Determine the algebraic and geometric multiplicities of each real eigenvalue. Determine whether the matrix is diagonalizable or not. You need not diagonalize it.

(a) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

(b) $B = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

(c) $C = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$.

like linear recurrence

Question 8. Consider Koch snowflake sequence:

$$0, 1, 13, 133, 1261, 11605, 105469, \dots$$

Here, the numbers can be described by a recursive formula,

$$K_{n+2} = 13K_{n+1} - 36K_n, \text{ where } K_0 = 0 \text{ and } K_1 = 1.$$

Let $\mathbf{u}_n = \begin{bmatrix} K_{n+1} \\ K_n \end{bmatrix}$.

(a) Find a matrix A such that $\mathbf{u}_{n+1} = A\mathbf{u}_n$.

(b) Let $\mathbf{u}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Prove, by induction, that $\mathbf{u}_n = A^n \mathbf{u}_0$ for all positive integers n .

(c) Diagonalise A and use it to prove that

$$K_n = \frac{1}{5}(9^n - 4^n).$$

Question 4(a) Set $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

What are the algebraic multiplicities of the real eigenvalues of A ?

characteristic polynomial: $\det(\hat{A} - \lambda \hat{I}) = \dots = \lambda(2 - \lambda)$

$$\lambda_1 = 0 \rightarrow \alpha(\lambda_1) = 1$$

$$\lambda_2 = 2 \rightarrow \alpha(\lambda_2) = 1$$

Question 4(a) Set $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

What are the **geometric** multiplicities of the real eigenvalues of A ?

$$\lambda_1 = 0 \quad 1 \leq \dim(E_0) \leq \alpha(\lambda_1) = 1 \Rightarrow \dim(E_0) = 1$$

$$\lambda_2 = 2 \quad 1 \leq \dim(E_2) \leq \alpha(\lambda_2) = 1 \Rightarrow \dim(E_2) = 1$$

$$\text{If } \alpha(\lambda) = 1 \Rightarrow \dim(E_\lambda) = 1$$

Question 4(a) Set $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

Is A diagonalizable?

$$\dim(E_{\lambda_i}) = \alpha(\lambda_i) \text{ for all } i$$

$$\sum \dim(E_{\lambda_i}) = n = 2$$

$\therefore A$ is diagonalizable

Alternatively, we say that $\hat{A}$ has $n=2$ distinct eigenvalues, immediately conclude that $\hat{A}$ is diagonalizable.

Question 4(b) Set $B = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

What are the algebraic multiplicities of the real eigenvalues of B ?

$$\text{Char poly } \det \begin{bmatrix} 2-\lambda & * & * \\ 0 & 3-\lambda & * \\ 0 & 0 & 3-\lambda \end{bmatrix} = (2-\lambda)(3-\lambda)^2$$

$$\lambda_1 = 2 \Rightarrow \alpha(\lambda_1) = 1$$

$$\lambda_2 = 3 \Rightarrow \alpha(\lambda_2) = 2$$

Question 4(b) Set $B = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

What is the **geometric** multiplicity of $\lambda_1 = 2$?

$$\lambda_1 = 2 \quad \alpha(\lambda_1) = 1 \Rightarrow \dim(E_{\lambda_1}) = 1$$

$$\left[\text{For } \lambda_1 = 2, \alpha(\lambda_1) = \dim(E_{\lambda_1}) \right]$$

alg = geom

Question 4(b) Set $B = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

What is the geometric multiplicity of $\lambda_2 = 3$?

$$\lambda_2 = 3$$

$$\alpha(\lambda_2) = 2$$

$$1 \leq \dim(E_3) \leq \alpha(3) = 2 \Rightarrow \text{cannot conclude}$$

Determine $E_3 = \text{null}(B - 3\hat{I}) = \text{null}\left(\begin{bmatrix} -1 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\right) \xrightarrow{\text{RREF}}$

$$\dim(\hat{E}_3) = 2$$

$\hookrightarrow$ we have 2 free parameters
 $\Rightarrow$ null space has dim 2.

Question 4(b) Set $B = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

Is B diagonalizable?

$$\alpha(2) = 1 = \dim(E_2)$$

$$\alpha(3) = 2 = \dim(E_3)$$

$$\dim(E_2) + \dim(E_3) = 3$$

$\therefore$ Yes. Diagonalizable.

Question 4(c) Set $C = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$.

What are the algebraic multiplicities of the real eigenvalues of C ?

char poly $(\lambda + 2)^2 (\lambda - 3)^2$

$$\lambda_1 = -2 \Rightarrow \alpha(-2) = 2$$

$$\lambda_2 = 3 \Rightarrow \alpha(3) = 2$$

Question 4(c) Set $C = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$.

What is the geometric multiplicity of $\lambda_2 = 3$?

$$E_3 = \text{null}(C - 3I) = \text{null}\left(\begin{bmatrix} -5 & 0 & 0 & 0 \\ 0 & -5 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\right)$$

$$\therefore \dim(E_3) = 1$$

$\uparrow 4 - 3 = 1$ free parameter

Question 4(c) Set $C = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix}$.

Is C diagonalizable?

No.

Because $\alpha(3) = 2 > \dim(E_3) = 1$

$\therefore C$ is NOT diagonalizable

Question 8. Consider the sequence

$$K_{n+2} = 13K_{n+1} - 36K_n, \quad K_0 = 0, K_1 = 1$$

Let $\mathbf{u}_n = \begin{bmatrix} K_{n+1} \\ K_n \end{bmatrix}$.

(a) Find A such that $\mathbf{u}_{n+1} = A\mathbf{u}_n$.

$$\mathbf{u}_{n+1} = \begin{bmatrix} K_{n+2} \\ K_{n+1} \end{bmatrix} = \begin{bmatrix} 13K_{n+1} - 36K_n \\ K_{n+1} \end{bmatrix}$$

$$\text{Let } \hat{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\hat{A}\mathbf{u}_n = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} K_{n+1} \\ K_n \end{bmatrix} = \begin{bmatrix} aK_{n+1} + bK_n \\ cK_{n+1} + dK_n \end{bmatrix} \quad \text{So, } \hat{A} = \begin{bmatrix} 13 & -36 \\ 1 & 0 \end{bmatrix}$$

Question 8. Consider the sequence

$$K_{n+2} = 13K_{n+1} - 36K_n, \quad K_0 = 0, K_1 = 1$$

Let $\mathbf{u}_n = \begin{bmatrix} K_{n+1} \\ K_n \end{bmatrix}$.

(b) Let $\mathbf{u}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Prove by induction that $\mathbf{u}_{n+1} = \mathbf{A}^n \mathbf{u}_0$ for all integers.

Question 8(c) Set $A = \begin{bmatrix} 13 & -36 \\ 1 & 0 \end{bmatrix}$.

Notice that

$$\left. \begin{aligned} \begin{bmatrix} 13 & -36 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} &= 4 \begin{bmatrix} 4 \\ 1 \end{bmatrix} \\ \begin{bmatrix} 13 & -36 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 9 \\ 1 \end{bmatrix} &= 9 \begin{bmatrix} 9 \\ 1 \end{bmatrix} \end{aligned} \right\} \text{found using characteristic polynomial}$$

Diagonalize A .

$$\hat{A} = \hat{P} \hat{D} \hat{P}^{-1} \quad \text{where} \quad \hat{P} = \begin{bmatrix} 4 & 9 \\ 1 & 1 \end{bmatrix}$$

$$\hat{D} = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}$$

Question 8(c)

So, $A = PDP^{-1}$.

What is A^n ?

$$\hat{A} = (\hat{P} \hat{D} \hat{P}^{-1})^n = \hat{P} \hat{D}^n \hat{P}^{-1}$$

$$= \hat{P} \begin{bmatrix} 4^n & 0 \\ 0 & 9^n \end{bmatrix} \hat{P}^{-1}$$

↳ what is $\hat{P}^{-1}$?

Question 8(c)

$$\text{So, } P = \begin{bmatrix} 4 & 9 \\ 1 & 1 \end{bmatrix}.$$

What is P^{-1} ?

$$\hat{P}^{-1} = -\frac{1}{5} \begin{bmatrix} 1 & -9 \\ -1 & 4 \end{bmatrix} \leftarrow \text{use standard } 2 \times 2 \text{ formula for inverse}$$

↓

$$\begin{aligned} A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} &\Rightarrow A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \\ &= \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \end{aligned}$$

Question 8(c) Put everything together.

$$A^n = P D^n P^{-1}$$

$$= \begin{bmatrix} 4 & 9 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4^n & 0 \\ 0 & 9^n \end{bmatrix} \left(-\frac{1}{5} \begin{bmatrix} 1 & -9 \\ -1 & 4 \end{bmatrix} \right)$$

$$= -\frac{1}{5} \begin{bmatrix} 4 & 9 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 4^n \mathbf{I} & 0 \\ 0 & 9^n \end{bmatrix} \begin{bmatrix} 1 & -9 \\ -1 & 4 \end{bmatrix}$$

$$= -\frac{1}{5} \left(4^n \begin{bmatrix} 4 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -9 \end{bmatrix} + 9^n \begin{bmatrix} 9 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 4 \end{bmatrix} \right)$$

Question 8(c) Almost there.

$$A^n = -\frac{1}{5} \left(4^n \begin{bmatrix} 4 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & -9 \end{bmatrix} + 9^n \begin{bmatrix} 9 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 4 \end{bmatrix} \right)$$

$$= -\frac{1}{5} \left(4^n \begin{bmatrix} 4 & -36 \\ 1 & -9 \end{bmatrix} + 9^n \begin{bmatrix} -9 & 36 \\ -1 & 4 \end{bmatrix} \right)$$

$$\text{So, } \hat{A} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = -\frac{1}{5} \left(4^n \begin{bmatrix} 4 \\ 1 \end{bmatrix} - 9^n \begin{bmatrix} -9 \\ -1 \end{bmatrix} \right) \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$
$$= \frac{1}{5} \begin{bmatrix} 9^{n+1} - 4^{n+1} \\ 9^n - 4^n \end{bmatrix}$$

$$\therefore K_n = \frac{1}{5} (9^n - 4^n)$$